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B. E. (Fifth Semester) Examination, April-May 2016

(New Scheme)

(Elect. Engg. Branch)

ELECTRICAL MACHINES-II

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Answer all questions. Part (a) from each unit is compulsory and carries 2 marks. Attempt any two parts from part (b), (c) and (d) of each question and carries 7 marks.*

Unit-I

1. (a) Write the conservation of energy in equation form. 2

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(b) Derive an expression for the torque in a doubly-excited system. 7

(c) A 3 phase, 50 Hz, 2 pole, star connected turbo alternator has 54 slots with 4 conductors per slot. The pitch of the coil is 2 slot less than pole pitch. If machine gives 3300 V between lines on open circuit with sinusoidal flux distribution, determine the useful flux per pole. 7

(d) Explain the phenomenon of MMF of concentrated windings and distributed windings. 7

Unit-II

2. (a) Give the reason for field winding to be on rotor in synchronous machine. 2

(b) Explain zero power factor method to determine voltage regulation of alternator. 7

(c) In a 50 kVA star connected 440 V, 3 phase 50 Hz alternator, the effective armature resistance is 0·25 Ω per phase. The synchronous reactance is 3·2 Ω per phase and leakage reactance is 0·5 Ω per phase.

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Determine at rated load and unity power factor : (i) internal emf, (ii) no load emf, (iii) percentage voltage regulation at full load. 7

(d) Explain SCR. Derive the equation in per unit (pu). 7

Unit-III

3. (a) What do you mean by stiffness of coupling? 2

(b) Two station generators A & B operate in parallel. Station capacity of A is 50 MW and that of B is 25 MW. Full load speed regulation of station A is 3% and full load regulation of station B is 3·5%. Calculate the load sharing if the connected load is 50 MW. No load frequency is 50 Hz. 7

(c) Derive and explain power angle characteristics of a Salient Pole Synchronous Machine. 7

(d) A 6600 V, 1200 kVa, 3 phase alternator is delivering full load at 0·8 power factor lagging. Its reactance is 25% and resistance negligible. By changing the excitation, the emf is increased by 30% at this load. Calculate the new values of current and power factor. The machine is connected to infinite bus bars. 7

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Unit-IV

4. (a) What is Slip Speed? 2

(b) Derive torque equations at standstill, at synchronous speed at condition for maximum torque. 7

(c) A 3 phase 50 Hz, 4 pole induction motor has a slip of 4%. Calculate (i) Speed of the motor (ii) Frequency of rotor emf. If the rotor has a resistance of 1 Ω and standstill reactance of 4 Ω, calculate power factor at (iii) stand still (iv) a speed of 1400 rpm. 7

(d) Explain torque slip characteristics of induction motor showing braking mode, motoring mode and generating mode. 7

Unit-V

5. (a) Why induction motor cannot run at synchronous speed? 2

(b) Describe the various methods of starting of induction motor in details. 7

(c) Discuss various methods to control the speed of induction motor. 7

(d) Explain Cogging and Crawling in induction motors in brief. 7

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